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KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Enabling agents to autonomously regulate knowledge utilization through situational self-awareness

Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge

Training Methods

Trajectory Imitation

ReAct

Directly imitates gold trajectories without situational judgment.

Trial-and-Error

Reflexion

Injects external feedback indiscriminately across all steps.

Knowledge-Augmented

KnowAgent, WKM

Floods every step with domain knowledge regardless of need.

Humans possess situational self-awareness — the metacognitive ability to assess whether they can handle a situation directly, need to pause and reflect, or must seek external help. KnowSelf brings this capability to LLM agents.

Three Situational Modes: Fast, Slow, and Knowledgeable Thinking



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

1

Self-Awareness Data Construction

Agent self-explores environments. At each step, a heuristic criterion classifies the situation into fast, slow, or knowledgeable thinking. Special tokens are inserted to mark each situation type.

2

Two-Stage Training

First, supervised fine-tuning (SFT) teaches the agent initial self-awareness patterns. Then, Reward-based Policy Optimization (RPO) further refines the agent's ability to generate appropriate special tokens.

3

Self-Aware Inference

During deployment, the agent generates special tokens that trigger either direct action, reflection, or knowledge retrieval from an external knowledge base.

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Task: Put a hot mug in cabinet

OpenAI-O1:

Incorrectly tries to take an egg from the microwave after opening it.

KnowSelf (Llama-8B):

Uses a Reflection token to realize it should heat the mug using the open microwave.

Task: Put a saltshaker in drawer

DeepSeek-R1:

Reflects that the drawer is open and tries to close it first (which fails).

KnowSelf:

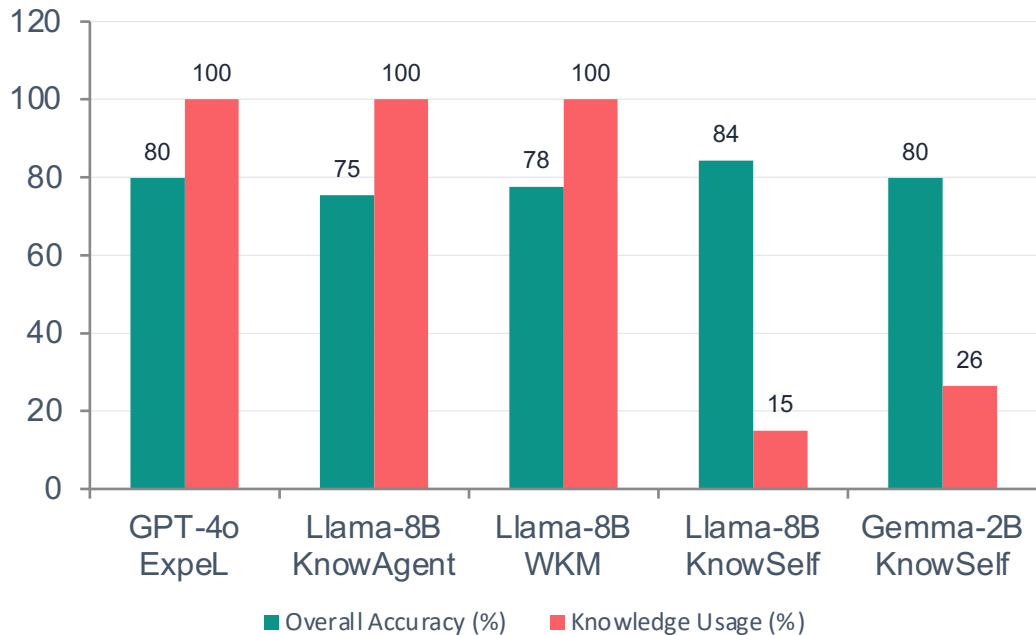
Applies a Knowledge token noting that the agent should place the object without removing unrelated items.

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Highlighted rows = KnowSelf. Teal Know% = selective knowledge. Red Know% = 100% knowledge injection.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage



Key Efficiency Insights

- KnowSelf (Llama-8B) achieves 84.33% overall — surpassing GPT-4o ExpeL (79.85%) — while using only 15% knowledge.
- Even Gemma-2B reaches 79.85% with KnowSelf, matching GPT-4o ExpeL and exceeding all other Gemma-2B baselines.
- Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform KnowSelf, confirming indiscriminate injection hurts performance.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

1. Indiscriminate injection is wasteful

Blanket knowledge use (100%) is suboptimal and incurs unnecessary inference cost.

2. Self-awareness is learnable

Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling.

3. KnowSelf outperforms strong baselines

Llama-8B KnowSelf (84.33%) beats GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%).

4. Only 15–26% knowledge needed

KnowSelf uses a fraction of the knowledge queries versus 100% for competing methods.

5. Scales across model sizes

Gemma-2B reaches 79.85%, matching GPT-4o ExpeL — demonstrating broad generalization.